

## 2018 JC2 PRELIMINARY EXAMINATIONS

CANDIDATE  
NAME

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INDEX NUMBER

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## H2 BIOLOGY

**9744/03**

**PAPER 3 Long Structured and Free-Response Questions**

**13 SEPT 2018  
THURSDAY**

**2 hours**

### READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.  
Write in dark blue or black pen.  
You may use an HB pencil for any diagrams or graph  
Do not use paper clips, highlighters, glue or correction fluid.

#### Section A

Answer **all** questions.

#### Section B

Answer any **one** question in the spaces provided on the Question Paper

| For Examiner's Use |      |
|--------------------|------|
| 1                  | / 26 |
| 2                  | /13  |
| 3                  | /11  |
| 4/5                | /25  |
| /75                |      |

The use of an approved scientific calculator is expected, where appropriate.  
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.  
The number of marks is given in brackets [ ] at the end of each question or part question.

**Section A**

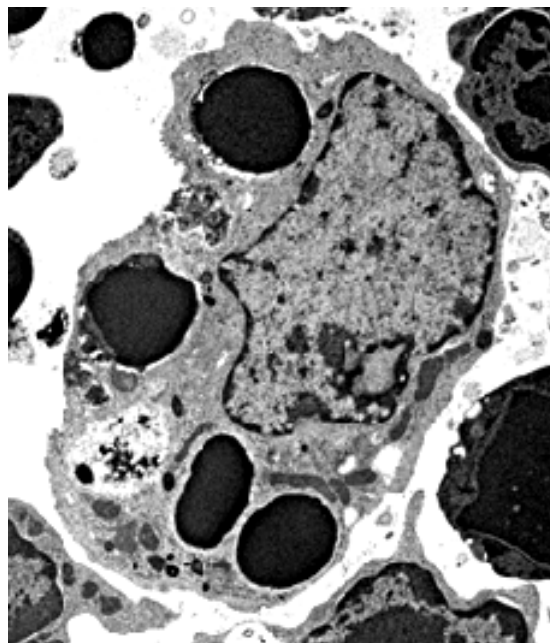
Answer **all** the questions in this section

- 1** Apoptosis is an important part of tissue homeostasis. It ensures that cells die quickly and are safely removed.

Every cell undergoing apoptosis shows a predictable series of changes. The cell shrinks and breaks up into dead cell fragments bounded by membranes. The membranes of these dead cell fragments display unique surface ligands, which identify the fragments to neighbouring cells ingesting these fragments.

Before the apoptotic cell breaks up, cellular DNA is degraded by endonucleases into small pieces that each differ in size by 180 base pairs.

Fig. 1.1 is an electron micrograph of a cell with ingested dead cell fragments.



**Fig. 1.1**

- (a) (i)** Identify **one** type of cell that can ingest dead cell fragments and explain how the cell recognises these cell fragments.

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[3]

- (ii) The DNA isolated from apoptotic cells can be analysed to confirm that apoptosis had indeed been activated.

Using the information above, outline a procedure that can be performed on a tissue extract to determine whether apoptosis has occurred.

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[3]

- (b) The understanding of apoptosis in human originated in the nematode worm *Caenorhabditis elegans*. Each worm is 1mm long and has a short life cycle. As the worm is transparent, scientists can follow how every cell in it originated from the zygote.

It was observed that in the formation of an adult worm, 1090 cells were generated, of which, exactly 131 cells died at particular points during the developmental process.

- (i) With reference to the information above, suggest why *C. elegans* is especially useful as an experimental organism.

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[2]

- (ii) A technique, known as the genetic screen, is used to identify genes that control a known process.

- For the case of *C.elegans*, scientists induced random genetic mutations by irradiating the eggs.
- The resulting adult worms were then examined.
- Analysis was then carried out to identify the genes that were being mutated.

In one of the genetic screen, it was observed that a particular *C.elegans* has more cells than normal as some cells did not die at particular points of the developmental process.

Suggest an explanation for the above observation.

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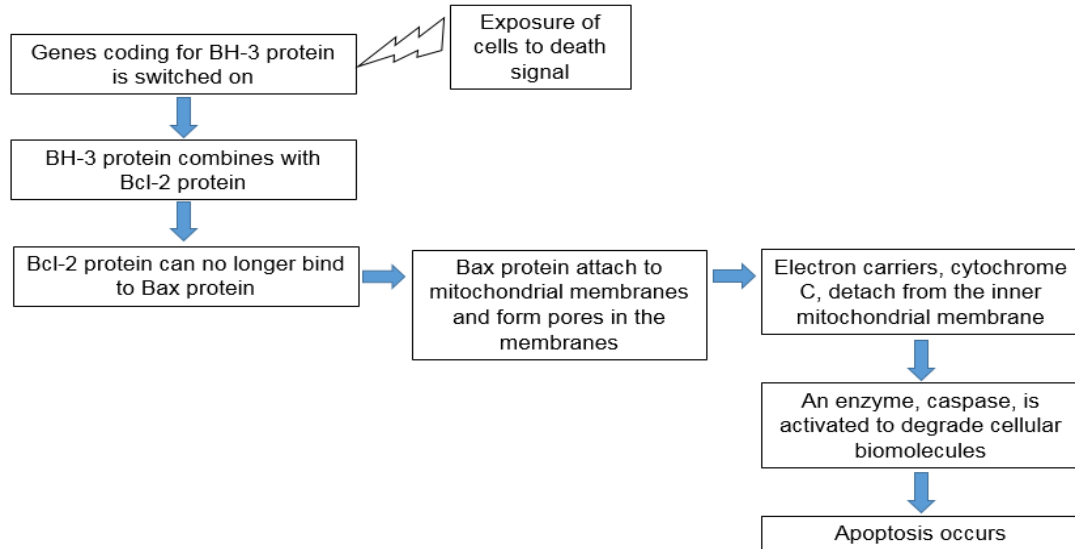
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[3]

(c) Fig. 1.2 shows one way in which apoptosis can be triggered.



**Fig.1.2**

- (i) With reference to Fig 1.2, predict and explain how ATP production will change when cytochrome C is detached from the inner mitochondrial membrane.

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[4]

- (ii) A particular mutation leads to the excessive production of Bcl-2 protein, disrupting the apoptotic pathway, leading to B-cell lymphoma, a cancer of B-cells.

- Bcl-2 gene is found on chromosome 18.
- In B-cell lymphoma, a mutation shifts the Bcl-2 gene to chromosome 14, bringing it close to a powerful gene regulatory sequence.
- This regulatory sequence regulates the production of the immunoglobulin heavy chain.

Explain why the mutation will lead to B-cell lymphoma.

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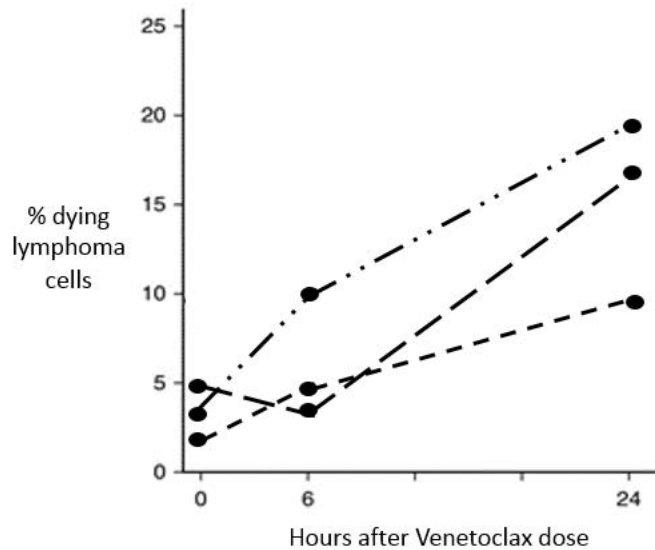
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[3]

- (d) A drug, Venetoclax has been developed recently to treat B-cell lymphoma. This drug is structurally similar to a BH-3 protein.

An experiment was performed on three different patients to investigate the effect of Venetoclax on lymphoma cells.

The results of the experiment are shown in Fig. 1.3.



**Fig 1.3**

With reference to Fig. 1.3 and the information above, describe and explain the effect of Venetoclax on cancer cells.

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(e) Justify the idea that apoptosis is a defence mechanism in humans.

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[Total: 26]

- 2 While modern therapeutic antibody production can involve other cell models, therapeutic antibodies are conventionally produced by animals (usually mice). For many specific antibodies, animal models remain the only viable option. Fig. 2.1 shows the simplified process of antibody production in rabbits.

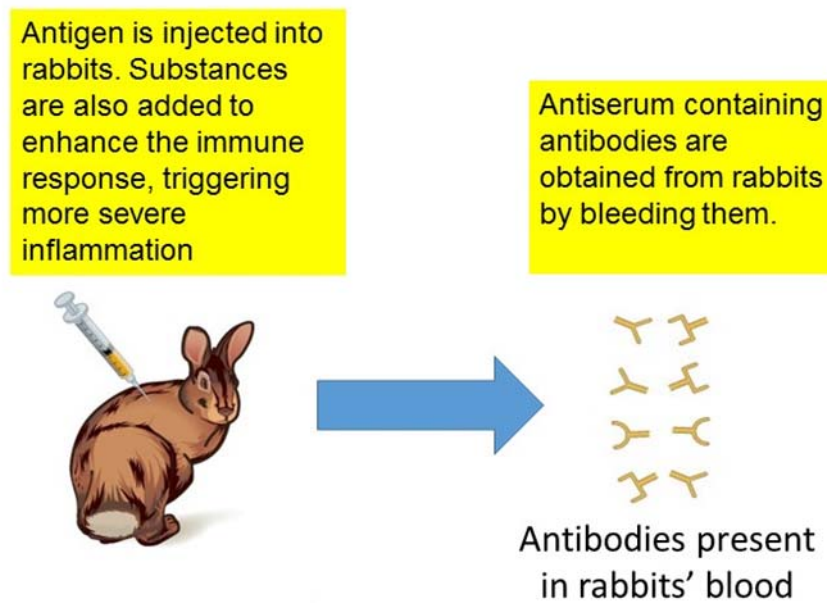


Fig. 2.1

- (a) Comment on the ethical implications of using animal models for large scale antibody production.

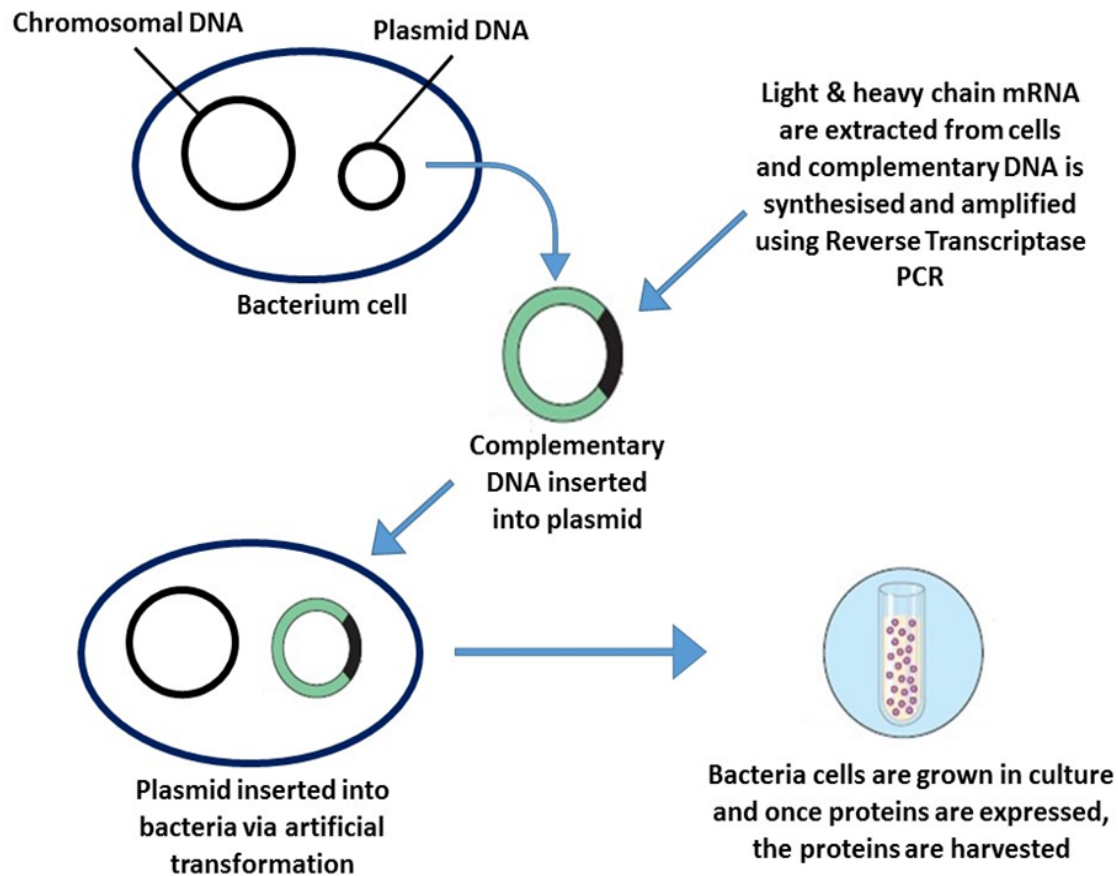
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Prokaryotic cells can potentially be used for antibody production. Fig. 2.2 shows a possible experimental procedure for antibody production in bacteria cells.



**Fig. 2.2**

To obtain DNA sequences of the light and heavy chain genes, light and heavy chain mRNA are extracted from plasma cells and then DNA containing the gene sequences are produced using reverse transcriptase Polymerase Chain Reaction (PCR).

The procedure for reverse transcriptase PCR is as follows:

- The extracted mRNA is mixed with DNA nucleotides, reverse transcriptase and incubated at 65°C for 5 minutes.
- RNase is then added to the reaction mixture and incubated at 42°C for 50 minutes, then 70°C for 15 minutes.
- Primers, *Taq* polymerase and DNA nucleotides are then added. The first cycle of PCR is carried out at 72°C, before the normal three-stage PCR cycle is carried out for the next 40 cycles at temperature 95°C, 60°C and 72°C.

**(b)** Explain the advantages of extracting mRNA instead of the DNA gene sequence from the genome of plasma cells.

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**(c)** For the reverse transcriptase PCR procedure, explain why

**(i)** extracted mRNA is mixed with DNA nucleotides and reverse transcriptase.

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[1]

**(ii)** RNase is added to the reaction mixture and eventually incubated at 70°C for 15 minutes.

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**(iii)** the first cycle of PCR is carried out at 72°C instead of the usual 95°C.

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[2]

**(d)** However, production of fully functional antibodies, which originate from eukaryotic cells, in prokaryotic cells is expected to be unsuccessful.

Explain why production of fully functional antibodies in prokaryotic cells is expected to be unsuccessful.

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[3]

[Total: 13]

- 3** To protect crops against insects, many crops such as maize and cotton are genetically modified to contain a gene (Bt) from a bacterium, *Bacillus thuringiensis*. The Bt gene codes for a toxin, which inserts into the insect gut cell membrane, paralyzing the digestive tract and killing the insects which eat the crop. When these genetically modified Bt crops first became available, it was predicted that insect pests would develop resistance to these toxins.

- (a)** Explain how Bt resistance may arise and spread in an insect population.

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[4]

- (b)** The extent of Bt resistance in insect pest species was surveyed in 2005 and in 2011. The level of resistance in each species was classified according to the highest percentage of resistant individuals recorded in any population anywhere in the world. Three levels of resistance were identified:

- <1%
- 1–6%
- >50%

There were no reports of populations of insect pests having between 6% and 50% of resistant individuals.

The results of the surveys are shown in Table 3.1.

**Table 3.1**

| year | total number of insect pest species surveyed | number of insect pest species susceptible to Bt toxins | number of insect pest species with reported levels of resistance |      |      |
|------|----------------------------------------------|--------------------------------------------------------|------------------------------------------------------------------|------|------|
|      |                                              |                                                        | <1%                                                              | 1–6% | >50% |
| 2005 | 9                                            | 8                                                      | 0                                                                | 0    | 1    |
| 2011 | 13                                           | 4                                                      | 3                                                                | 1    | 5    |

- (i) Discuss whether the data provided can support the hypothesis that rise in Bt resistance is due to introduction of Bt crops, including limitations of the data.

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- (ii) Justify the prediction that both insect-caused and non-insect-caused damages to Bt crops will increase with climate change.

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[Total: 11]

## Section B

Answer **one** question in this section

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answer must be set out in parts **(a)**, **(b)**, etc., as indicated in the question.

- 4 (a) Explain how fluidity of biological membranes can be maintained and the importance of fluidity to membrane function. [10]
- (b) A human body can synthesise more than 90,000 different types of proteins. The complete set of proteins present in the various cells of an individual is known as the proteome.

A human's genome is constant, whereas the proteome of a human is constantly changing.' Discuss this statement. [15]

[Total:25]

- 5 (a) Explain why more energy is released from the oxidation of triglycerides as compared to the same unit mass of glucose and outline the problems of using triglycerides as the main respiratory substrate instead of glucose in animals. [10]
- (b) Discuss the inter-relationships of membranous organelles in eukaryotic cells and outline the implication of absence of membranous organelles in prokaryotic cells. [15]

[Total: 25]

[illegible]







